CHA-2.B.2 The derivative of f is the function whose value at x is $\lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$, provided this limit exists.
CHA-2.B.3 For $y = f(x)$, notations for the derivative include dy/dx, $f'(x)$, and y'.
CHA-2.B.4 The derivative can be represented graphically, numerically, analytically, and verbally.
CHA-2.C.1 The derivative of a function at a point is the slope of the line tangent to a graph of the function at that point.

2-2 Definition of the Derivative and Derivative Notation

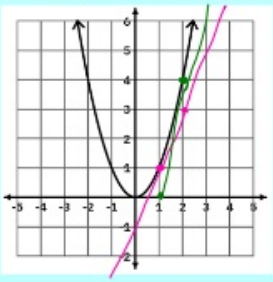
History of Calculus: Newton vs. Leibniz

In the history of calculus, the calculus controversy was an argument between mathematicians Isaac Newton and Gottfried Wilhelm Leibniz over who had first discovered calculus. Leibniz had published his work on calculus first, but Newton's supporters accused Leibniz of plagiarizing Newton's unpublished ideas. Newton stated he had begun working on a form of calculus (which he called "The Method of Fluxions and Infinite Series") in 1666, at the age of 23, but did not publish it until 1737 as a minor annotation in the back of one of his works decades later. Gottfried Leibniz began working on his variant of calculus in 1674, and in 1684 published his first paper employing it, "Nova Methodus pro Maximis et Minimis". L'Hôpital published a text on Leibniz's calculus in 1696 (in which he recognized that Newton's Principia of 1687 was "nearly all about this calculus"). Meanwhile, Newton, though he explained his (geometrical) form of calculus in Section I of Book I of the Principia of 1687, did not explain his eventual fluxional notation for the calculus in print until 1693 (in part) and 1704 (in full). The modern consensus is that the two men independently developed their ideas. [1]

[1] Wikipedia Leibniz-Newton Calculus Controversy

The derivative is an expression that calculates the instantaneous rate of change (slope of the tangent line) of a function at any given x-value. In other words, it gives us the slope of the function at a point.

$f(x) = x^2$



$f'(x) = 2x$

$f'(1) =$ $2(1)=2$

$f'(2) =$ $2(2)=4$

$f'(-2) =$ $2(-2)=-4$

Notation for the derivative of a function	**Lagrange Notation**	**Leibniz Notation**
	First derivative: $f'(x)$ or y'	First derivative: $\frac{dy}{dx}$ This means "the derivative of y with respect to x."

Definition of the Derivative of $f(x)$	The <u>derivative of $f(x)$</u>, denoted as $\boldsymbol{f'(x)}$, is $$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$ This limit gives an expression that calculates the instantaneous rate of change (or slope of the tangent line) of $f(x)$ at any <u>given x</u>-value.

Example 1: Given $f(x) = x^3 - x$

(a) Use the definition of the derivative to find $f'(x)$ for the given function

(b) Find $f'(2)$.

ⓐ $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \to 0} \frac{(x+h)^3 - (x+h) - (x^3 - x)}{h}$

$= \lim_{h \to 0} \frac{\cancel{x^3} + 3x^2h + 3xh^2 + h^3 - \cancel{x} - h - \cancel{x^3} + \cancel{x}}{h}$

$= \lim_{h \to 0} \frac{3x^2h + 3xh^2 + h^3 - h}{h}$

$= \lim_{h \to 0} (3x^2 + 3xh + h^2 - 1)$

$= 3x^2 - 1$

ⓑ $f'(2) = 3(2)^2 - 1 = 11$

Equation of a tangent line	The line tangent to the curve of $f(x)$ at $x = a$ can be represented in point-slope form: $$y - y_1 = m(x - x_1)$$ where $(x_1,\ y_1)$ is $(a, f(a))$ and $m = f'(a)$

Example 2: Given $f(x) = \sqrt{x-1}$

(a) Use the definition of the derivative to find $f'(x)$ for the given function

(b) Find $f'(5)$.

(c) Find the equation of the line tangent to the graph of $f(x)$ at $x = 5$

ⓐ $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \to 0} \frac{\left(\sqrt{x+h-1} - \sqrt{x-1}\right)}{h} \cdot \frac{\left(\sqrt{x+h-1} + \sqrt{x-1}\right)}{\left(\sqrt{x+h-1} + \sqrt{x-1}\right)}$

$= \lim_{h \to 0} \frac{x+h-1 + (-x+1)}{h\left(\sqrt{x+h-1} + \sqrt{x-1}\right)}$

$= \lim_{h \to 0} \frac{h}{h\left(\sqrt{x+h-1} + \sqrt{x-1}\right)}$

$= \lim_{h \to 0} \frac{1}{\sqrt{x+h-1} + \sqrt{x-1}}$

$= \frac{1}{\sqrt{x-1} + \sqrt{x-1}} = \frac{1}{2\sqrt{x-1}}$

ⓑ $f'(5) = \frac{1}{2\sqrt{5-1}} = \frac{1}{4}$

ⓒ slope $= f'(5) = \frac{1}{4}$ at $x=5$ $f(5) = \sqrt{5-1} = 2$

EQ_{TAN}: $y - 2 = \frac{1}{4}(x-5)$